Koi Pond Kit

Your own interactive koi pond. Put it on your website, then make every koi, ripple and colour yours, by hand or with AI.

- No installs, no coding needed
- Works on any website
- 3 looks built in
- Written for your AI too



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MAKE IT YOURS
Colours, ripples, koi
Playground and every setting · 5

USE AI
Describe it, get it
Prompts and a map for your AI · 13

HI, AND THANK YOU :)

I made this pond for the home page of my own site, and so many of you asked how it's done that I've packaged it up so you can have one too.

You don't need to know how to code. Everything you'd want to change has a name and a number, there's a playground with sliders, and if you'd rather just describe what you want, part of this guide is written for your AI to read.

Have fun with it. I can't wait to see your ponds.

Vivian

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WHAT'S IN THE ZIP

`index.html`
Your pond page, ready to publish. The ✏️ comments mark what to change: the words, your links and your look.

`playground.html`
Sliders and colour pickers for everything. Find your look, press “Copy my settings”, paste into index.html.

`koi-pond.js`
The pond itself: koi, water, ripples, pads. Every setting and its default is at the top, in plain words.

`koi-pond.css`
The soft dark edges of the pond and the little caption pills that follow the link koi.

`examples/react/`
A `<KoiPond>` component for React and Next.js, in TypeScript and JavaScript.

`AGENTS.md`
A map of the code for AI helpers. Coding tools like Cursor pick it up by themselves.

`README.md`
The two-minute version of this guide.

`LICENSE.txt`
Use it on your sites and client sites; please don't resell or share the code.

Works everywhere
Chrome, Safari, Edge and Firefox, on computers, phones and tablets.

Light on phones
Fewer koi and pads on small screens, and it rests when scrolled out of view.

Kind to everyone
Links work with a keyboard and screen readers, and it calms down for people who prefer less motion.

01 QUICK START

Your pond, with your name on it

About ten minutes. All you need is a browser and a text editor.

1 Unzip and open `index.html`

Double-click it and it opens in your browser. Tap the water, drag through it, and hover over a koi with a caption. Keep all the files together in one folder, always.

2 Open it in a text editor

VS Code is free and colours the code so it's easier to read (code.visualstudio.com). TextEdit on a Mac works too: Format → Make Plain Text first. On Windows, Notepad is fine. Tip: copy the whole folder before you start, so you always have the original.

3 Put your words in

Find the ✏️ comments. The words in the middle are near the bottom of the file:

```
<h1 class="title">
  <span class="hi reveal" style="--i: 0">hi, i'm</span>
  <span class="name reveal" style="--i: 1">your name</span>
</h1>
<p class="subtitle reveal" style="--i: 2">welcome to my little pond :)</p>
```

Change the browser tab's name in `<title>` at the top too. The fonts, sizes and the pink shimmer on your name are in the `<style>` block under "The words in the middle".

4 Set your links

Each link is a koi with a caption that follows it around. Click it and the pond splashes, then off you go.

```
const links = [
  { label: "my work", href: "#work", koi: "gold", x: 0.36,
    tint: "#F0BC55", ink: "#FBE9C4" },
  { label: "say hi", href: "mailto:you@example.com", koi: "platinum", x: 0.64,
    tint: "#E6ECF6", ink: "#EEF1F6" },
];
```

`label · href`
The caption, and where it goes: a page, a full web address, or `mailto:`.

`koi · x`
Which koi carries it (page 10) and where it likes to swim, 0 left to 1 right.

`tint · ink`
The caption's colour and its text colour. `[]` means no link koi at all.

5 Save, then refresh the browser

That's the loop for everything in this guide: change, save, refresh. Next, put it online, or jump to the playground to change how it looks.

Typing code by hand?
Keep the quotes, commas and brackets exactly as they are. If the pond disappears after a change, undo it (Cmd/Ctrl + Z), save, and try again.

02 PUT IT ONLINE

Pick the way that fits your site

Plain files, so it works with every host and site builder. Always upload all of them together.

A · The pond is your whole website

Easiest · free

1 · Netlify Drop

Go to **app.netlify.com/drop** and drag the whole **koi-pond-kit** folder onto the page.

2 · Your link

You get a web address straight away, like `calm-koi-123.netlify.app`.

3 · Keep it

Make a free account to keep the site, rename it, and add your own domain.

Vercel, GitHub Pages and Cloudflare Pages work the same way: upload the folder.

B · Inside Squarespace, Wix, Framer, Webflow, Shopify, Carrd…

Any site builder

Site builders don't let you upload script files, so host the pond first (A above), then show it on your page with an embed. Add your builder's embed or code block and paste this in, with your own address:

```
<iframe src="https://calm-koi-123.netlify.app" title="Koi pond"
  style="display:block;width:100%;height:100vh;border:0"></iframe>
```

Where's the embed block?

Squarespace: Code block · Wix: Embed code → Embed HTML · Framer: Embed · Webflow: Embed · Shopify: Custom Liquid section · Carrd: Embed. Change `100vh` for a shorter pond, like `600px`.

Two settings for embedded ponds

Add `target: "_top"` to each link, so it opens the whole page and not just the pond. And if the pond is one section of a longer page, set `allowScroll: true` in your look so phones can scroll past it.

C · Your own HTML site

Copy `koi-pond.js` and `koi-pond.css` next to your page and add:

```
<link rel="stylesheet" href="koi-pond.css">

<div id="pond" style="height: 100vh">
  <!-- anything here floats on the water;
       give it position: relative; z-index: 3
-->
</div>

<script src="koi-pond.js">
```

D · React and Next.js

Put `koi-pond.js` and `koi-pond.css` in `public/`, copy `examples/react/KoiPond.tsx` (or `.jsx`) into your components, and load `/koi-pond.css` in your layout.

```
"use client";
import { useRouter } from "next/navigation";
import KoiPond from "@/components/KoiPond";

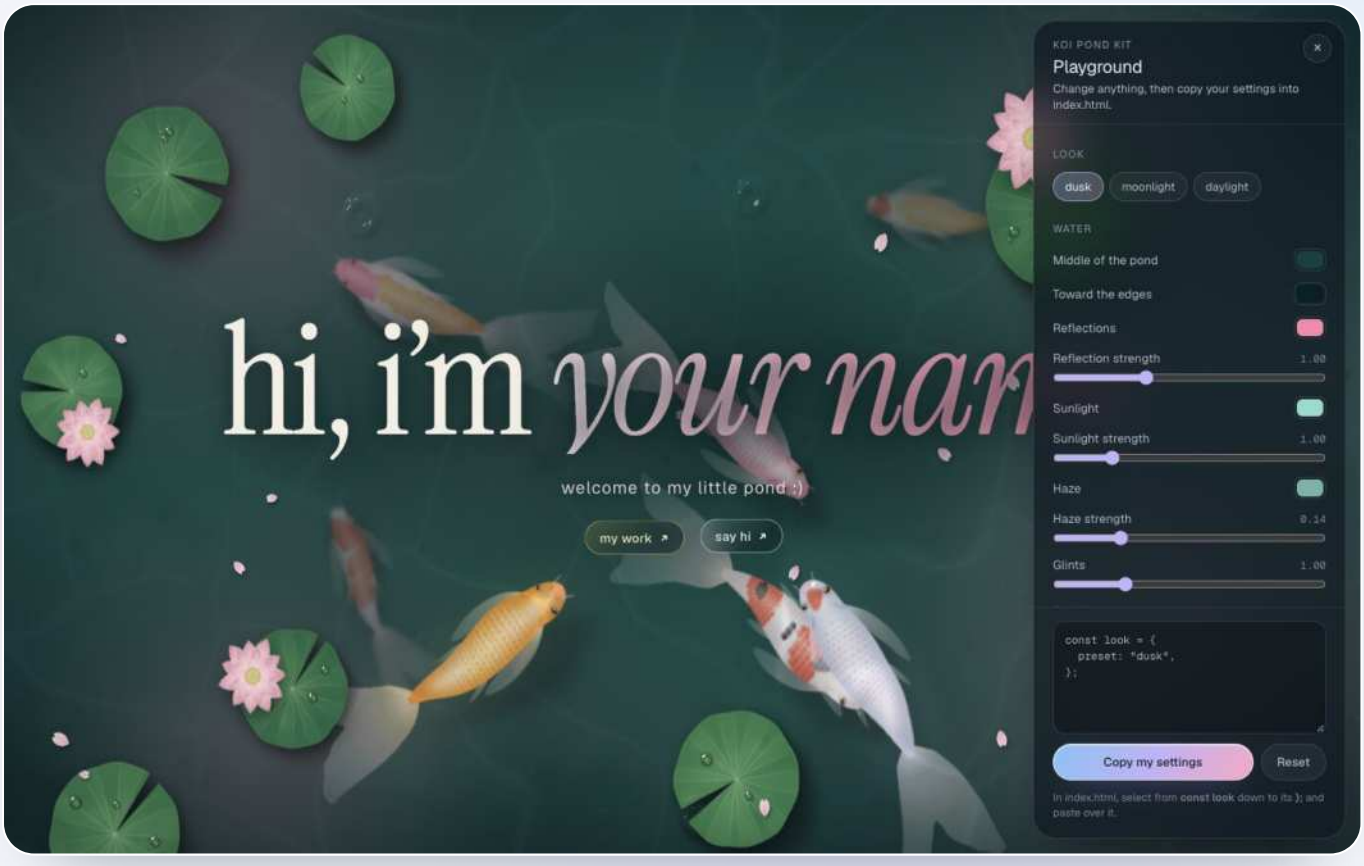
export default function Home() {
  const router = useRouter();
  return (
    <KoiPond settings={{ ...look, links,
      navigate: (href) => router.push(href),
    }}>
      <h1 className="title">hi, i'm …</h1>
    </KoiPond>
  );
}
```

`look` and `links` are the same as in index.html. Vue, Svelte, Astro: call `KoiPond.create(el, settings)` when the component mounts and `pond.destroy()` when it leaves.

03 THE PLAYGROUND

Slide until it feels right

Open `playground.html`. Every change shows straight away, and the box at the bottom writes your settings for you.



1 Pick a look to start from

Dusk, moonlight or daylight. Switching starts that look fresh.

2 Change anything

Water colours and light, how ripples behave, which koi swim and how, pads and flowers. Koi and pad changes re-stock the pond, so they appear somewhere new.

3 Press “Copy my settings”

You get a block that starts with `const look = {`. It only lists what you changed.

4 Paste it into index.html

Select from `const look = {` down to its `};` and paste over it. Save, refresh. Your words and links stay as they were.

What you copy looks like this — paste over the old block

```
const look = {
  preset: "moonlight",
  water: {
    glow: "#FFD27A",
    glowAmount: 2.2,
  },
  koi: {
    types: ["sakura", "platinum", "showa"],
  },
  ripples: {
    fade: 0.993,
  },
};
```

Lost the panel?

The × hides it so you can see the whole pond; “Tweak the pond” brings it back. On phones it sits along the bottom.

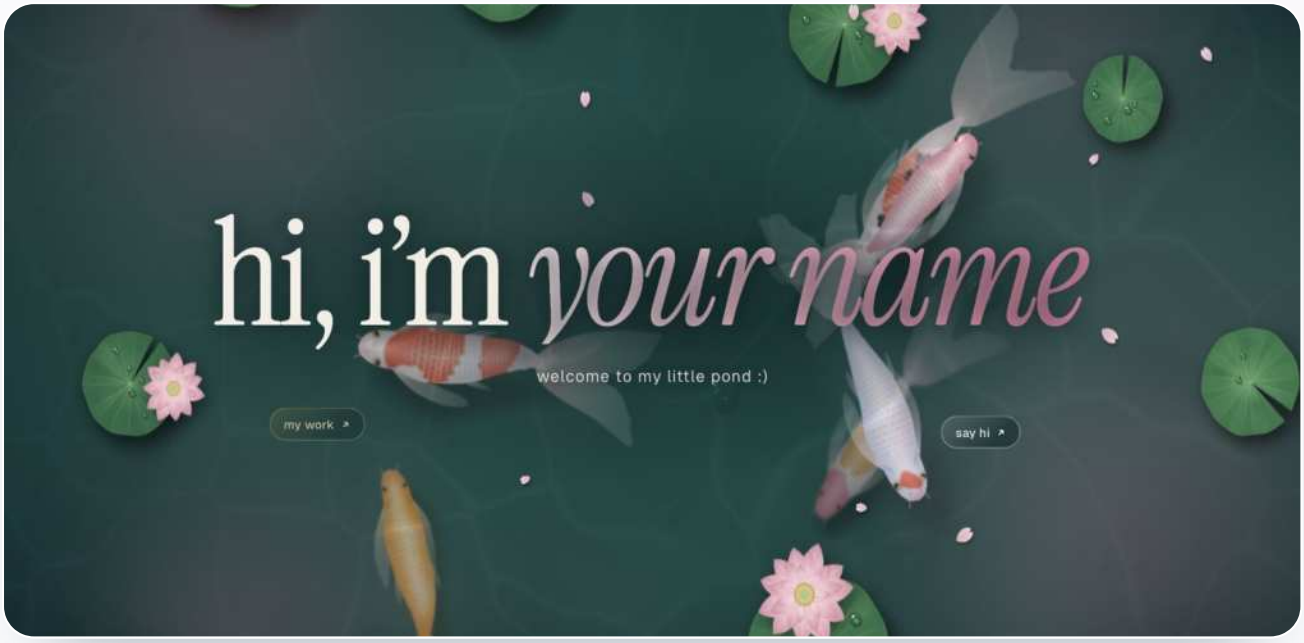
The playground is just for you

Your website doesn't need `playground.html`. Anything the sliders don't cover is in the next chapters, and all of it can go in your `look`.

04 THREE LOOKS

Dusk, moonlight and daylight

Each preset sets the water, light, koi, lotuses and petals in one word. Start from the closest one, then change what you like.



Dusk the default

Deep green water, pink evening light, sakura and kohaku koi. The pond from my site.

`preset: "dusk"`



Moonlight `"moonlight"`

Inky blue water, a moon in the corner, silver and black koi, white lotuses.



Daylight `"daylight"`

Clear jade water, bright sunlight on the floor, classic red, orange and black koi.

Mix a preset with your own changes

```
const look = {
  preset: "moonlight",
  water: { glow: "#F6B3C6" }, // a pink moon
  koi: { types: ["gold", "platinum"] },
};
```

Yours wins

Anything you set is used over the preset's. Anything you leave out comes from the preset.

Make your own preset

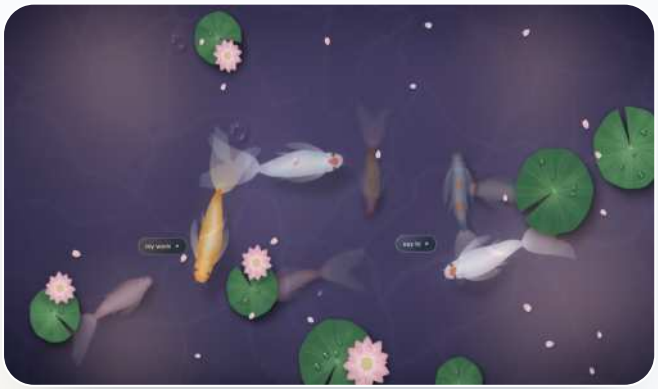
Presets live in section 3 of `koi-pond.js`. Add one there and you can switch to it with one word, like `preset: "sakura"`.

05 COLOURS OF EVERYTHING

Every colour, and where it lives

Colours are hex codes like `"#F08CAB"`, with the `#` and the quotes. The defaults below are dusk's. Most go in your `look`; a few are in the page's CSS.

WHAT	SETTING	DEFAULT	NOTES
WATER · IN YOUR LOOK			
Open water	`water.middle`	#1C403D	Lighter in the middle of the pond…
Toward the banks	`water.edges`	#0B2023	…darker toward the edges
Before it loads	`water.background`	#12302F	Shows for a blink; match `theme-color` in index.html
Milky haze	`water.haze` · `hazeAmount`	#80B3A8 · 0.14	Soft clouds drifting through the water, 0 to 1
Sunlight net	`water.light` · `lightAmount`	#9EDBCC · 1	The wobbly web of light on the floor
Reflections	`water.glow` · `glowAmount`	#F08CAB · 1	Sky light in the water; where it sits is `water.glows` ↓
Surface sheen	`water.sheen` · `sheenAmount`	#FFDBE6 · 1	A faint wash of sky across the surface
Glints	`water.sparkle` · `sparkleAmount`	#FFEDF2 · 1	Sun catching the ripples
The deep	`water.tint` · `sink`	#10292A · 1	What koi fade into as they swim down; `sink: 0` keeps them bright
PADS, FLOWERS, PETALS · IN YOUR LOOK			
Lily pads	`pads.colors`		A list; each pad picks one. Plain hex works: `["#4E8A4F", "#5A9553"]`
Lotus	`lotus.petal` · `blush` · `tip`		From the middle of each petal out to its tip
Lotus centre	`lotus.centre` · `stamens` · `pod` · `seeds`		The golden middle, its ring, the seed pod and its seeds
Fallen petals	`petals.colors` · `vein`		Base to tip, and the line down the middle
Koi	`koi.types`	Their own chapter: page 10	
WORDS AND CAPTIONS · IN INDEX.HTML AND KOI-POND.CSS			
Title and name	`.title` · `.name`		In index.html's `<style>`; the name's shimmer is its `linear-gradient`
Link captions	`links[].tint` · `ink`		Per link. Shape and glass are `.koi-pond__link` in koi-pond.css
Dark edges	`.koi-pond::after`		The vignette, in koi-pond.css. Delete that block for none



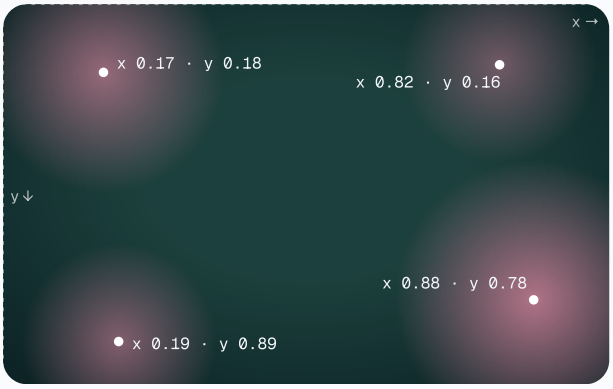
A lavender evening, made only with the settings on the right.

```
const look = {
  water: {
    middle: "#3B3558", edges: "#14112A",
    background: "#221E3C", haze: "#BDB2E6",
    light: "#DCD0FF", glow: "#F9C6A8",
    glowAmount: 1.3, sheen: "#F2E8FF",
    tint: "#1B1730",
  },
  koi: { types: ["blush", "platinum",
    "sakura", "asagi"] },
  petals: { count: 22 },
};
```

05 COLOURS OF EVERYTHING

Moving the light around

The pink glow in the water isn't one colour wash: it's up to six soft pools of light you can place, size and dim.



Dusk's four reflections. x and y go from 0 (left, top) to 1 (right, bottom). **size** is a share of the pond's height, **strength** how bright it is.

```
const look = {
  water: {
    glow: "#FFE7A8", // warm sunlight
    glows: [
      // a big pool, top right
      { x: 0.85, y: 0.12,
        size: 0.5, strength: 0.25 },
      // a faint one, bottom left
      { x: 0.1, y: 0.9,
        size: 0.3, strength: 0.06 },
    ],
  },
};
```

A list replaces the whole list

Setting `glows` swaps out all of the preset's pools, so list every one you want. `glows: []` turns them all off; so does `glowAmount: 0`.

Recipes

Sakura night

pink everything

```
water: { middle: "#3B2A3A", edges: "#170F1C",
  background: "#2A1D29", tint: "#24161F",
  glow: "#FFB3C7", glowAmount: 1.4,
  light: "#FFD6E2" },
koi: { types: ["sakura", "blush", "platinum"] },
petals: { count: 40, countOnPhones: 20 },
```

Clear mountain pool

see right to the bottom

```
preset: "daylight",
water: { middle: "#4F9C93", edges: "#1F5E5E",
  hazeAmount: 0.05, lightAmount: 3, sink: 0.4 },
pads: { count: 3, countOnPhones: 2 },
petals: { count: 0, countOnPhones: 0 },
```

Golden hour

warm and hazy

```
water: { middle: "#3A4A35", edges: "#141C12",
  haze: "#E8C99A", hazeAmount: 0.25,
  light: "#FFE0A8", lightAmount: 1.6,
  glow: "#FFB46B", sheen: "#FFE2C2" },
```

Your brand colours

swap in your own hex codes

```
water: { middle: "#__", edges: "#__",
  glow: "#__" },
lotus: { blush: "#__", tip: "#__" },
petals: { colors: ["#__", "#__"] },
koi: { types: [{ base: "#__", spots: [
  { color: "#__", count: 3, size: 0.6 }] }] },
```

Each recipe goes inside your `const look = { … };`. Links go in `const links`.

06 RIPPLES & MOVEMENT

How the water feels to touch

Taps splash, drags cut a wake, koi near the surface stir it, and the odd raindrop lands on its own. All of it is in `ripples`.



SETTING	DEFAULT	FEELS LIKE
`ripples.fade`	0.982	How long ripples last. 0.95 settles almost at once; 0.99 rings for ages. Keep it under 1, or they'd never stop.
`ripples.bend`	1	How strongly ripples bend the floor and the koi under them, like a lens. 2 to 3 is dramatic.
`ripples.tap` · `tapSize`	240 · 4	The splash from a tap or click: how hard, and how wide. `tap: 0` for none.
`ripples.drag`	1	The wake when you drag a finger or the mouse through the water. 0 for none.
`ripples.wake`	6	The little ripples koi leave as they swim near the surface.
`ripples.rain` · `rainEvery` · `rainStrength`	true · [0.9, 2.4] · 70	Drops landing on their own, every 0.9 to 2.4 seconds.
`water.swell`	1	Slow, gentle waves over the whole pond. 0 makes it glassy.
`pads.push`	true	The pointer nudges pads and petals along as it passes.
`koi.shy`	true	Koi dart away from the pointer.

Glassy and calm

```
ripples: { rain: false,
  wake: 2 },
water: { swell: 0.2 },
```

Rainy day

```
ripples: {
  rainEvery: [0.1, 0.4],
  rainStrength: 110 },
```

Big splashes

```
ripples: { tap: 600,
  tapSize: 7 },
```

Long, silky ripples

```
ripples: { fade: 0.992,
  bend: 1.6 },
```

Look, don't touch

```
ripples: { tap: 0, drag: 0 },
koi: { shy: false },
pads: { push: false },
```

Choppy

```
ripples: { fade: 0.97,
  wake: 14 },
water: { swell: 2.2 },
```

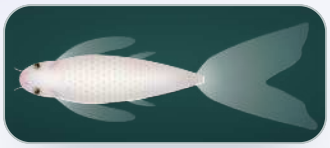
Calmer for people who ask for it

If someone's device is set to reduce motion, the koi swim slower and the rain and koi wakes stop. Taps still splash.

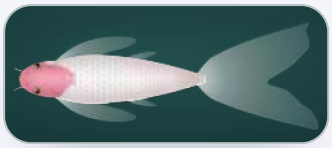
07 REDESIGN THE KOI

A koi is a little recipe

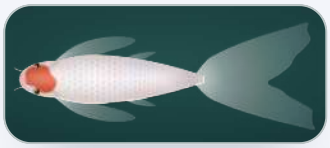
Start with a body colour, then add any of these. Every koi is painted fresh when the pond loads, so patches land a little differently each time, like real koi.



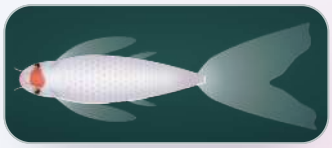
base body colour



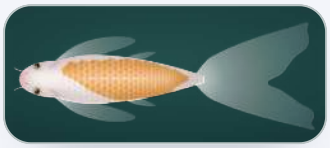
head over the head



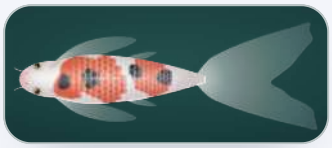
cap top of the head



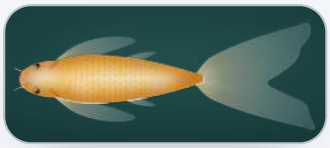
dot spot on the crown



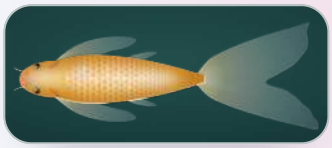
saddle band down the back



spots any number of patches



sheen: 0 matte



sheen: 1 metallic

```
{
  base: "#F8F3EE", // the only one you need
  head: "#F28CA4",
  cap: "#E4502F",
  dot: "#E4502F",
  saddle: "#F2A447",
  spots: [
    // big red patches, then little ink spots
    { color: "#EE6A4D", count: 3, size: 1 },
    { color: "#2C2A30", count: 4, size: 0.3 },
  ],
  sheen: 0.6, // 0 matte … 1 metallic
  fin: "#FCD6DE", // optional: fins
  shade: "#D6C4CE", // optional: sides
  eye: "#CFA45C", // optional: iris
}
```

Patches stack in order

Saddle first, then spots, then head, cap and dot on top.

fin and shade fill themselves in

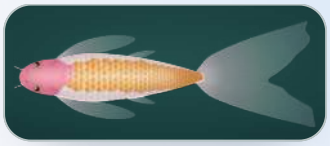
Leave them out and they're worked out from `base`. Set them when a dark or very bright koi needs a hand.

spots is a list

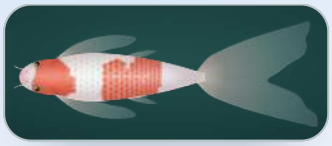
Each entry is one colour: how many patches (`count`) and how big (`size` , 1 is normal).

The koi in the kit

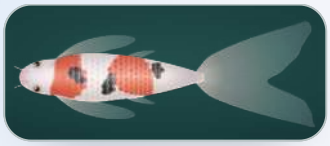
Use these names in `koi.types` and in each link's `koi` .



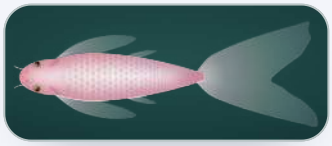
sakura dusk, moonlight



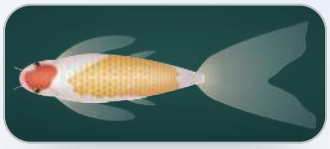
kohaku all three



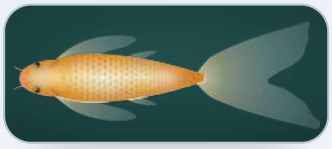
sanke dusk, daylight



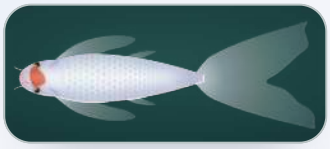
blush dusk



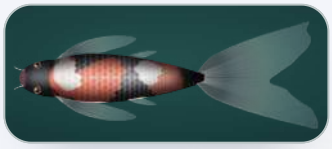
sunrise dusk



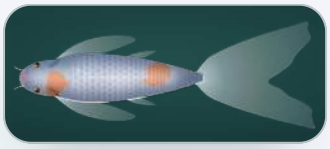
gold link koi, daylight



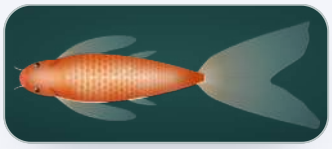
platinum link koi, moonlight



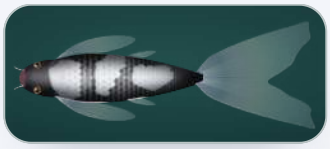
showa daylight



asagi moonlight



tangerine daylight



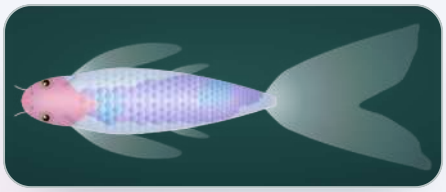
midnight moonlight

Yours here

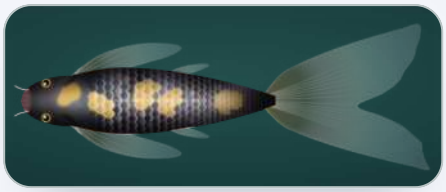
Add your own to `KOI_TYPES` in section 1 of koi-pond.js to use it by name.

07 REDESIGN THE KOI

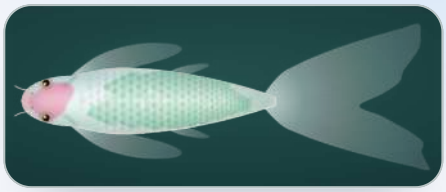
Your own koi, how they swim, and their shape



Pastel sky, lilac, blush



Ink & gold black with gold leaf



Mint with a pink cap

Write them straight into your look — names and your own koi can mix

```
const look = {
  koi: {
    types: [
      "kohaku",
      { base: "#E4F0FD", saddle: "#BFB4F5", head: "#F4A9C4", sheen: 1,
        spots: [{ color: "#8EC1F2", count: 2, size: 0.5 }] },
      { base: "#1F1E24", fin: "#6B5A3A", eye: "#E2B145", sheen: 1,
        spots: [{ color: "#E2B145", count: 5, size: 0.42 }] },
      { base: "#E5F7EF", saddle: "#A8E2C8", cap: "#F4A9C4", sheen: 0.9 },
    ],
  },
};
```

Koi take the types in turn: with 7 koi and 4 types, the first three types get two koi each. A link koi can be your own too: `koi: { base: "#E4F0FD", … }`.

How they swim

SETTING	DEFAULT	WHAT IT DOES
`koi.count`	7	Koi besides the link koi
`koi.countOnPhones`	4	Ponds under 700px wide
`koi.size`	0.24	Length, as a share of the pond's shorter side
`koi.minSize` · `maxSize`	100 · 230	Never smaller or bigger than this, in px
`koi.speed`	[24, 44]	px per second; each koi picks one
`koi.wiggle`	1	Swish of body and tail
`koi.wander`	1	Meandering; 0 swims long smooth arcs
`koi.dive`	true	Rise to the surface and sink again
`koi.shy`	true	Dart away from the pointer
`keepCenterClear`	true	Stay out of the middle, around the title

Their shape in koi-pond.js

Shape is shared by every koi, so it lives in section 5 of `koi-pond.js`, not in your look. Search for the name:

`WIDTH = 0.13`
How wide at the widest. 0.16 is chubby, 0.1 slender.

`TAIL = 0.62`
Tail length. 0.9 or so for long, flowing butterfly koi.

`PEC_SHAPE · PEL_SHAPE`
Outlines of the front and back fins. Easiest to ask your AI: “make the front fins 40% longer”.

`paintTail → tipU · tipD · fork`
Where the tail's tips and fork sit. Bigger numbers, wider tail.

Faces, scales and whiskers

All painted in `paintBody`: look for “Lips, nostrils and eyes” and “Whiskers”. Scales come from `scaleGrid`. These are the best things to hand to your AI with a description.

What floats on top, and the koi that carry links

Lily pads, lotuses and petals

SETTING	DEFAULT	WHAT IT DOES
`pads.count` · `countOnPhones`	7 · 4	How many pads
`pads.size`	[42, 86]	Radius in px (phones: `sizeOnPhones`)
`pads.lotusEvery`	2	A lotus on every 2nd pad. 1 for all, 0 for none
`pads.drops`	true	Beads of water on the leaves
`pads.drift`	1	How fast they float about
`pads.push`	true	The pointer pushes pads and petals
`petals.count` · `countOnPhones`	14 · 8	Fallen petals drifting on top
`petals.size`	[6, 11]	px

Colours are on page 7. Pads drift off one edge and come back on the other, so the pond never empties.

Link koi

FIELD	EXAMPLE	WHAT IT DOES
`label`	"my work"	The caption
`href`	"/work"	Where it goes
`koi`	"gold"	Its look: a name, or your own
`x` · `y`	0.36 · 0.74	Where it likes to swim (0 to 1)
`tint`	"#F0BC55"	Caption colour and the koi's glow
`ink`	"#FBE9C4"	Caption text colour
`target`	"_blank"	New tab; `"_top"` inside an iframe

Hover a caption and its koi slows down so it's easy to catch. Two to four links look best; keep their `x` apart.

Three links, one opening in a new tab

```
const links = [
  { label: "about", href: "/about",
    koi: "platinum", x: 0.25 },
  { label: "work", href: "/work",
    koi: "gold", x: 0.5 },
  { label: "shop", koi: "sakura", x: 0.75,
    href: "https://shop.example.com",
    tint: "#F4A9C4", ink: "#FDE9F1",
    target: "_blank" },
];
```

Caption style — koi-pond.css

```
.koi-pond__link {
  padding: 9px 15px; /* pill size */
  border-radius: 999px; /* 8px: soft */
  font-size: 13.5px;
  background: linear-gradient(135deg,
    rgba(var(--tint), 0.16),
    rgba(18, 40, 42, 0.6) 70%);
  border: 1px solid rgba(var(--tint), 0.5);
  color: var(--ink);
}
```

No title in the middle?

Set `keepCenterClear: false` and koi and pads use the whole pond.

A pond partway down the page

Give its element a height (like `600px`) and set `allowScroll: true` so phones can scroll past it.

09 CUSTOMISE WITH AI

Describe it, and let your AI write it

The kit is built to be easy for AI to change: one file, named sections, and a map written just for it.

Chat AI · ChatGPT, Claude, Gemini

Start a new chat and attach **koi-pond.js**, **AGENTS.md** and **index.html**. Add this PDF too if it reads PDFs. Paste the prompt below and describe what you want. Copy back what it gives you.

Coding AI · Cursor, Copilot, Claude Code, Codex

Open the **koi-pond-kit** folder. Most of these read **AGENTS.md** by themselves; if not, say “read AGENTS.md first”. They edit the files for you, so just refresh the browser.

The starter prompt — copy it, then add your wish at the end

> I'm customising the Koi Pond Kit, an interactive koi pond for my website. I've attached koi-pond.js, AGENTS.md and index.html; please read AGENTS.md first. I'm not a developer, so: change settings in index.html when you can, and koi-pond.js only when you have to; tell me exactly which file and which part to replace; give me the complete code to paste, not just the changed lines; and keep koi-pond.js a plain script that still works when I open index.html from my computer. Here's what I'd like: …

Ideas to try — one at a time works best

- “Make every koi black with small gold flecks, like ink and gold leaf.”
- “Give the koi long, flowing butterfly tails and bigger front fins.”
- “Make the water soft lavender with peach reflections, like the sky after sunset.”
- “Make ripples bigger and slower to fade, like a stone dropped in.”
- “When I tap the water, drop a pink petal where I tapped.”
- “Add a third link koi called “shop” that opens my store in a new tab.”
- “Switch to moonlight after 7pm and back to dusk in the morning.”
- “Make the koi swim toward the pointer, like it's feeding time.”
- “Make the pond a 600px section on my page that phones can scroll past.”
- “Make the link captions solid white pills with dark text.”
- “Make one koi in my brand colours: #… and #…”
- “Put my logo in the middle of the pond instead of the title.”

One change at a time

Check it, then ask for the next.

Keep a copy

Duplicate the folder before big changes.

Something broke?

Right click → Inspect → Console. Paste any red text to your AI.

Ask why

“Explain what you changed” teaches you the code as you go.

10 FOR YOUR AI

How the kit works

Written for AI helpers, so it's technical: skip it, or hand it over. AGENTS.md in the zip says the same.

For AI The user may not be a developer. Prefer settings over code edits, name the exact file and section, and return complete code to paste.

Architecture

`koi-pond.js` is one classic script (an IIFE that sets `window.KoiPond`), plain JS, no dependencies or build. **Keep it that way:** no `import` / `export` or npm; it must work when index.html is opened from disk. The water and koi are WebGL2 (GLSL ES 3.00), drawn in two passes: koi skins as triangle strips into an offscreen layer (`MESH_*`), then a full-screen `WATER` shader for floor, caustics, koi, reflections and surface. Pads, lotuses and petals go on a 2D canvas above. Link captions are real `<a>` elements. No WebGL2 → a plainer 2D fallback.

Settings

Merged `DEFAULTS` → `PRESETS[preset]` → user settings. Objects merge per key; **arrays replace.** Colours must be hex strings (others render magenta on purpose). In index.html the user's settings are `const links = […]` and `const look = {…}`, passed as `{ ...look, links }`. New setting: add it to `DEFAULTS` with a comment, read `opts.group.name`; for `update()` to apply it without re-stocking, add its path to `LIVE`. New shader value: add the uniform, list it in `WATER_SETTINGS`, set it in `applyLook`.

Map of koi-pond.js search for the heading, e.g. "// 5. KOI SKINS"

SECTION	HOLDS	EDIT IT TO…
`1. KOI TYPES`	`KOI_TYPES`, colour constants	add or change koi looks
`2. SETTINGS`	`DEFAULTS`	change defaults, add settings
`3. PRESETS`	`PRESETS`	add a named look
`4. HELPERS`	`mix`, `merge`, `resolve`, `koiType`, `blob`	patch shapes (`blob`), colour maths
`5. KOI SKINS`	`WIDTH`, `TAIL`, `PEC_SHAPE`, `PEL_SHAPE`, `scaleGrid`, `paintBody`, `paintTail`, `paintKoi`	outline, tail, fins, scales, patches, face. Atlas 1024×512: body and tail in the top 400px, fins below
`6. SURFACE`	`padSprite`, `paintLotus`, `paintPetal`, `paintGlow`, `paintDrop`	leaves, lotus rings, petal shape, water beads
`7. POND FLOOR`	`paintFloor`	silt and stones (greys; the shader tints them)
`8. WATER`	`WATER` (`caustics`, `swell`, `glow`, `main`), `MESH_FRAGMENT`, `setupGl`, `MAX_GLOWS`	the water's look; how koi fade with depth
`9. KOI BODIES`	`shapeOf`, `bodyPath`, `along`	how the body bends; shadow outline
`10. THE POND`	`create()`: `applyLook`, `makeKoi`, `spawn`, `swim`, `ripple`, `disturb`, `drawTop`, `render`, `buildLinks`, `onMove`, `loop`	behaviour: steering (`swim`), ripple physics (a damped wave equation on a 3px grid, `CELL`), pointer effects, fin beat, drifting, captions
`11. EXPORTS`	`window.KoiPond`	add public helpers

API

```
const pond = KoiPond.create(element, settings);
pond.update({ water: { glow: "#FFD27A" } });
pond.splash(x, y, strength); // px, top left
pond.settings;   // merged, in use
pond.destroy();  // removes it all
KoiPond.defaults · .presets · .types
KoiPond.paintKoi(type); // a skin canvas
KoiPond.resolve(settings);
```

Layers inside the element

Water canvas z 0 · pads and petals canvas z 1 · vignette `::after` z 2 · page content z 3 · link captions and hit areas z 4. The element needs a height.

Keep working

Opening from disk · `prefers-reduced-motion` (`reduced`) · phone counts · keyboard-usable links · `destroy()` undoing all of `create()`. Don't shrink `CELL`: the ripple grid is the heaviest part.

10 FOR YOUR AI

Every setting

All optional. Defaults are dusk's; moonlight and daylight change the water, koi types, pads, lotus and petal colours.

SETTING	DEFAULT
`preset`	"dusk" · "moonlight" · "daylight"
KOI	
`count` · `countOnPhones`	7 · 4 (phones: under 700px wide)
`types`	["sakura", "kohaku", "sanke", "blush", "sunrise"]
`size`	0.24 of the shorter side
`minSize` · `maxSize`	100 · 230 px
`speed`	[24, 44] px/s
`wiggle` · `wander`	1 · 1 (multipliers)
`dive` · `shy`	true · true
LINKS (LIST)	
`label` · `href`	caption · destination
`koi`	"gold" (name or object)
`x` · `y`	spread evenly · 0.74
`tint` · `ink`	"#F0BC55" · "#FFFFFF"
`target`	none · "_blank" · "_top"
RIPPLES	
`fade`	0.982 (clamped 0.8–0.998)
`bend`	1
`tap` · `tapSize`	240 · 4 (cells)
`drag` · `wake`	1 · 6
`rain` · `rainEvery` · `rainStrength`	true · [0.9, 2.4] s · 70
PAGE	
`keepCenterClear`	true
`allowScroll`	false (true: touch-action pan-y)
`navigate`	null · function(href)

SETTING	DEFAULT
WATER	
`middle` · `edges`	"#1C403D" · "#0B2023"
`background`	"#12302F"
`haze` · `hazeAmount`	"#80B3A8" · 0.14
`light` · `lightAmount`	"#9EDBCC" · 1
`glow` · `glowAmount`	"#F08CAB" · 1
`glows`	4 × { x, y, size, strength } (max 6)
`sheen` · `sheenAmount`	"#FFDBE6" · 1
`sparkle` · `sparkleAmount`	"#FFEDF2" · 1
`swell`	1
`tint` · `sink`	"#10292A" · 1
PADS	
`count` · `countOnPhones`	7 · 4
`size` · `sizeOnPhones`	[42, 86] · [28, 52] px radius
`colors`	3 × { base, light, dark, vein } or hex
`lotusEvery`	2 (0 none)
`drops` · `drift` · `push`	true · 1 · true
LOTUS	
`petal` · `blush` · `tip`	"#F6ECEA" · "#E9C4CD" · "#C98D9F"
`centre` · `stamens`	"#DCC489" · "#C9A56A"
`pod` · `seeds`	"#BFCB90" · "#8E9E66"
PETALS	
`count` · `countOnPhones`	14 · 8
`size`	[6, 11] px
`colors` · `vein`	["#DDB4BF", "#ECD2D8"] · "#C495A3"

Live in update()

`water`, `ripples`, `koi.wiggle`, `wander`, `dive`, `shy`, `pads.push`, `keepCenterClear`, `allowScroll`, `navigate`.

Re-stocks the pond

Anything else in `update()` makes new koi and pads. Skins are cached per type, so unchanged koi don't repaint.

Koi type fields

`base` (required), `head`, `cap`, `saddle`, `dot`, `spots[{color,count,size}]`, `sheen`, `shade`, `fin`, `eye`.

11 TROUBLESHOOTING

If something's not right

Most problems are a missing file or a missing comma. The browser's console (right click → Inspect → Console) usually says which.

The page is blank, or just dark green

Keep `koi-pond.js` and `koi-pond.css` in the same folder as `index.html`, with those exact names. If you just edited `index.html`, a quote or comma may be missing: undo and try again.

It works on my computer but not online

Upload every file, not just `index.html`. Online, file names are case-sensitive: `Koi-Pond.js` isn't `koi-pond.js`.

My change didn't show up

Save the file, then refresh. Still the same? Force a fresh copy: Cmd + Shift + R on a Mac, Ctrl + F5 on Windows.

Something turned bright magenta

That's a colour that isn't a hex code. Write it like `"#F08CAB"`, with the `#` and the quotes.

I can't scroll past the pond on my phone

Set `allowScroll: true` in your look. The pond takes touches by default so you can drag through the water.

A link opens inside the pond's little window

The pond is in an iframe: add `target: "_top"` to that link.

It's slow on an old phone

Lower `koi.countOnPhones`, `pads.countOnPhones` and `petals.countOnPhones`. A shorter pond is lighter too.

The water looks flat, with no light or ripples

That browser has WebGL2 switched off, so the plain backup pond shows. Updating the browser, or turning on hardware acceleration in its settings, usually fixes it.

The fonts look different

`index.html` loads Geist and Instrument Serif from Google Fonts, so you need to be online. Change them in the `<link>` at the top and in the `<style>` block.

Koi keep swimming over my title

Make sure `keepCenterClear` is `true`. They still cross now and then, which is part of the charm; the darker water behind the words keeps them readable.

What you can do with it

Use it on as many of your own sites as you like, personal or business, and on sites you make for clients. Change it however you like. Please don't resell, share or give away the kit or its code, changed or not, and don't say you made it. Full terms in `LICENSE.txt`.

Stuck?

Try the starter prompt on page 13 with the console message. Still stuck? Message me wherever you found the kit and I'll help where I can.

THAT'S EVERYTHING

Have fun in the water :)

Vivian

I'd love to see what you make with it.